XIAOYU ZHANG

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Professional Experience

City University of Hong Kong, School of Creative Media Aug. 2024 - Present
- Tenure-track Assistant Professor

ETH Zurich, ETH AI Center Aug. 2023 - July 2024
- Postdoctoral fellow, mentored by Prof. Mrinmaya Sachan and Prof. Torbjørn Netland
- Topic: Human-in-the-loop visual knowledge discovery and human-centered AI

Bosch Research, Visual Analytics & eXplainable AI Group Sep. 2022 - June 2023
- Research Intern, supervised by Dr. Liu Ren, Dr. Jorge Piazentin Ono, and Dr. Wenbin He
- Topic: A Visual Concept-Based Slice Finding Approach for Image Data

Meta, Data Infrastructure Group May 2022 - Aug 2022
- Research Data Scientist Intern, supervised by Dr. Jian Shen
- Topic: Stacktrace Theme Analysis and Anomaly Detection with NLP Techniques

Bosch Research, Visual Analytics & eXplainable AI Group June 2021 - May 2022
- Research Intern, supervised by Dr. Liu Ren, Dr. Liang Gou, and Dr. Jorge Piazentin Ono
- Topic: A Data Slice-Driven Approach for Machine Learning Model Validation

Microsoft Research Asia, Visual Computing Group May 2018 - Aug. 2018
- Research Intern, supervised by Dr. Weiwei Cui
- Topic: Automatic Generation of Infographics from Natural Language Statements

NetEase Games, Game Engine Group Jan. 2017 - Mar. 2017
- Senior Research & Development Intern
- Topic: Customized Character Face Creation in MMORPG Games based on Skeleton-Driven Animation

University of Alberta, Surgical Simulation Research Lab Mar. 2016 - Sep. 2016
- Visiting Scholar, supervised by Prof. Bin Zheng
- Topic: Define Interior Structure for Better Liver Segmentation Based on CT Images

Hong Kong University, Department of Computer Science July 2014 - Sep. 2014
- Research Assistant, supervised by Prof. Wenping Wang
- Topic: Automated Steady Scaffoldings for 3D Printing

Education

Ph.D. Degree: Computer Science, University of California, Davis, USA Sep. 2018 - June 2023
- Research interests: Visual Analytics, Information Visualization, Human-Computer Interaction
- Advisor: Prof. Kwan-Liu Ma

Master's Degree: Computer Science, Zhejiang University, China Sep. 2015 - Mar. 2018
- Research interests: Computer Graphics, Medical Image Processing
- Advisor: Prof. Kun Zhou, Prof. Zhong Ren

Bachelor's Degree: Digital Media Art, Xiamen University, China Sep. 2011 - Jun. 2015
- Major: Digital Media Art (GPA: 3.86 / 4.0)
- Minor: Advertising (GPA: 3.58 / 4.0)

Research Interests

I am broadly interested in research topics related to **data visualization, human-computer interaction (HCI) and human-AI collaboration**. Specifically, I contribute interactive tools that incorporate data visualization techniques with state-of-the-art machine learning techniques to support knowledge exploration and exploitation for **education, cultural heritage, and human well-being**.

Publications

Journal Publications

[J1] **Xiaoyu Zhang**, Vincent Dörig, Peng Cui, Vilém Zouhar, Torbjörn Netland, Mrinmaya Sachan : *Harmonizing Assistance: Moderating Visual and Textual Aids in AI-Enhanced Textbook Reading with IRead*, International Journal of Artificial Intelligence in Education (IJAIED), 2025.

[J2] **Xiaoyu Zhang**, Jorge Henrique Ono, Huan Song, Liang Gou, Kwan-Liu Ma, and Liu Ren: *SliceTeller: A Data Slice-Driven Approach for Machine Learning Model Validation*, IEEE Transactions on Visualization and Computer Graphics (2022).
Best Paper Honorable Mentioned Award on IEEE VIS 2022

[J3] Xiwei Xuan, **Xiaoyu Zhang**, Oh-Hyun Kwon, and Kwan-Liu Ma: *VAC-CNN: A Visual Analytics System for Comparative Studies of Deep Convolutional Neural Networks*, IEEE Transactions on Visualization and Computer Graphics 28, no. 6 (2022): 2326-2337.

[J4] Weiwei Cui, **Xiaoyu Zhang**, Yun Wang, He Huang, Bei Chen, Lei Fang, Haidong Zhang, Jian-Guan Lou, and Dongmei Zhang: *Text-to-Viz: Automatic Generation of Infographics from Proportion-Related Natural Language Statements*, IEEE Transactions on Visualization and Computer Graphics 26, no. 1 (2019): 906-916.

Peer-Reviewed Conference Publications

[C1] Zhao, Zijian, Dian Jin, Zijing Zhou, **Xiaoyu Zhang**: *Automatic stage lighting control: Is it a rule-driven process or generative task?*, In International Conference on Learning Representations (ICLR), vol. 2026, pp. 136041-136064. 2026.

[C2] Yao, Rui, Qiuyuan Ren, Felicia Fang-Yi Tan, Yang Chen, **Xiaoyu Zhang**, Shengdong Zhao: *PersonaMail: Learning and Adapting Personal Communication Preferences for Context-Aware Email Writing*, In Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI), pp. 197-217. 2026.

[C3] **Xiaoyu Zhang**, Fei Xue, Alexander Albers, Torbjørn Netland: *"It's impressive, but in practice...": Experiencing a Realistic Digital Transformation in and beyond the Classroom*, In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems.

[C4] Li Nan, Fan Shi, **Xiaoyu Zhang**, April Wang: *MRISA: A Visual Analytics Approach for Comparing Locomotion Policies in Robotics Training*, In 51st Graphics Interface Conference (GI'25), 2025.

[C5] Lennart Wittke, Peter El Hachem, Yuan Chen, Joseph Ollier, Bingjie Cheng, Shkurta Gashi, **Xiaoyu Zhang**: *Classifying Cognitive Load in Digital Textbook Reading using Electroencephalogram Signals*, In Proceedings of the 13th International Conference on Information and Education Technology (ICIET), 2025.
Best Presentation Award

[C6] Peng Cui, Vilém Zouhar, **Xiaoyu Zhang**, Mrinmaya Sachan: *How to Engage Your Readers? Generating Guiding Questions to Promote Active Reading*, In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024.

[C7] **Xiaoyu Zhang**, Jorge Henrique Ono, Wenbin He, Liang Gou, Mrinmaya Sachan, Kwan-Liu Ma, and Liu Ren: *Slicing, Chatting, and Refining: A Concept-Based Approach for Machine Learning Model Validation with ConceptSlicer*, In 29th Annual ACM Conference on Intelligent User Interfaces (IUI), 2024.

[C8] **Xiaoyu Zhang**, Kelvin Li, Po-Wei Chi, Senthil Chandrasegaran, and Kwan-Liu Ma: *ConceptEVA: Concept-Based Interactive Exploration and Customization of Document Summaries*, In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems.

[C9] **Xiaoyu Zhang**, Xiwei Xuan, Alden Dima, Thurston Sexton, and Kwan-Liu Ma: *LabelVizier: Interactive Validation and Relabeling for Technical Text Annotations from Weak Supervision*, In 2023 IEEE 16th Pacific Visualization Symposium (PacificVis).

[C10] **Xiaoyu Zhang**, Senthil Chandrasegaran, and Kwan-Liu Ma: *Conceptscope: Organizing and visualizing knowledge in documents based on domain ontology*, In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (pp. 1-13).

[C11] **Xiaoyu Zhang**, Takanori Fujiwara, Senthil Chandrasegaran, Michael P. Brundage, Thurston Sexton, Alden Dima, and Kwan-Liu Ma: *A Visual Analytics Approach for the Diagnosis of Heterogeneous and Multidimensional Machine Maintenance Data*, In 2021 IEEE 14th Pacific Visualization Symposium (PacificVis) (pp. 196-205).

[C12] Senthil Chandrasegaran, **Xiaoyu Zhang**, Michael P. Brundage, and Kwan-Liu Ma: *Using Text Visualization to Aid Analysis of Machine Maintenance Logs*, Model-Based Enterprise (MBE) Summit 2020.

[C13] **Xiaoyu Zhang**, Yixiong Zheng, Bin Zheng: *Define Interior Structure for Better Liver Segmentation Based on CT Images*, In CCF Chinese Conference on Computer Vision, pp. 77-88. Springer, Singapore, 2017.

[C14] Yingying She, Qian Wang, **Xiaoyu Zhang**, Wenyong Zheng, Yao Fu, Yishi Shi: *The Approach of Micro-blog Explosive Events Detection and Analysis in Real-time*, International Conference on Computer Engineering and Information Systems. Atlantis Press, 2016.

Posters, Demos and Workshops

[W1] Jiayi Sun, **Xiaoyu Zhang**: *Blooming Vowels: Visualizing Four Hu and Emotional Expression in Mandopop Lyrics*, In Proceedings of the IEEE VIS Poster Track, 2026.

[W2] Guo, Yuanlei, Xizi Gong, Yizhong Zhang, **Xiaoyu Zhang**: *Magical Touch: Transforming Raw Capacitive Streams into Expressive Hand-Touchscreen Interaction*, In Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems, pp. 1-5. 2026.

[W3] Yao, Rui, Ruiyuan Bai, Yang Chen, Qiuyuan Ren, Felicia Fang-Yi Tan, **Xiaoyu Zhang**, Shengdong Zhao: *Demonstrating PersonaMail: Learning and Adapting Personal Communication Preferences for Context-Aware Email Writing*, In Companion Proceedings of the 31st International Conference on Intelligent User Interfaces, pp. 160-166. 2026.

[W4] Daria Martynova, Jakub Macina, Nico Daheim, Özge Nilay Yalçın, **Xiaoyu Zhang**, Mrinmaya Sachan: *Can LLMs Effectively Simulate Human Learners? Teachers' Insights from Tutoring LLM Students*, In Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA), 2025.

[W5] Qingxiaoyang Zhu, **Xiaoyu Zhang**, Shuyan Dai, Noriko Satake, and Hao-Chuan Wang: *ZoombaTogether: A Video Conference Add-on for Generating Interactive Visual Feedback for Online Group Exercise Through Real-Time Pose Tracking*, ACM SIGCHI Conference on Designing Interactive Systems (DIS) 2023.

Creative and Literary Works

[V1] Weiqi Han, Yupu Chen, Yunzhong Luo, **Xiaoyu Zhang**: *Hidden Information Behind Fancy Words: A Comparative Data-Driven Analysis of ESG Reports*, 2026 Visual Data Storytelling Contest (In conjunction with IEEE PacificVis). 🏆 **Honorable Mentioned Award**

[V2] Guo, Yuanlei, Xizi Gong, Yizhong Zhang, **Xiaoyu Zhang**: *A Cat's Day*, 2026 Visual Data Storytelling Contest (In conjunction with IEEE PacificVis 2026).

[V3] Jonghoon Daniel Kim, Woojang Pyeon, **Xiaoyu Zhang**: *South Korea's Untold Stories: Unveiling Birth Policies through Data Visualization*, 2025 Visual Data Storytelling Contest (In conjunction with IEEE PacificVis 2025).

Patent & Software Copyrights

[P1] Jorge Henrique Ono, **Xiaoyu Zhang**, Liang Gou, Liu Ren: System and method for efficient analyzing and comparing slice-based machine learn models. US Patent (US20240135160A1).

[P2] Jorge Henrique Ono, **Xiaoyu Zhang**, Huan Song, Liang Gou, Liu Ren: System and method for a visual analytics framework for slice-based machine learn models. US Patent (US20240135159A1).

[P3] Yingying She, **Xiaoyu Zhang**, Wenyong Zheng, Yao Fu, Yishi Shi: Micro-blog Explosive Events Detection and Analysis System V1.0. Xiamen University. Registration No.: 2015SR218554.

[P4] Yingying She, **Xiaoyu Zhang**, Wenyong Zheng, Yao Fu, Yishi Shi: Micro-blog Explosive Events Detection and Analysis System (iOS Version) V1.0. Xiamen University. Registration No.: 2015SR218546.

Grants

[G1] *Constructing a Reusable Knowledge Base and Immersive Learning Experience for Hong Kong Martial Arts Archives*. Principal Investigator / Project Coordinator, Donation, 9220206, $200,000.

[G2] *Classifying cognitive load using textbook reading in older adults with an ECG-based machine learning approach*.
Principal Investigator / Project Coordinator, City University of Hong Kong and University of Waterloo, 9667276, HKD 150,000 + CAD 26,000.

[G3] *Visualizing and Structuring Semantic Information in Machine Maintenance Logs based on Domain Ontology*.
Contributor (PI: Kwan-Liu Ma), National Institute of Standards and Technology, 2021-NIST-MSE-01, $89,505.00.
Proposed research ideas and wrote the majority of the proposal.

[G4] *Visualization-Guided Technical Language Processing for Machine Maintenance Logs*.
Contributor ((PI: Kwan-Liu Ma), National Institute of Standards and Technology, 2020-NIST-MSE-01, $89,505.00.
Provided some initial work on which the project is based, detailed development and evaluation plan.

Teaching

Instructor

- SM3801 Understanding Data (Undergraduate) — Spring 2026
- SM6351 Information Visualization (Graduate) — Fall 2025
- SM3749 Spec. Topics in Creative Media (Undergraduate) — Fall 2025
- SM2705 Creative Coding (Undergraduate) — Spring 2025, 26
- SM1103 Introduction to Media Computing (Undergraduate) — Fall 2024
- 263-5051-00L AI Center Projects in Machine Learning Research (Graduate) — Spring 2024
- ECS 32B Introduction to Data Structures (Undergraduate) — Summer 2022

Guest Lecturer

- ECS 273 Visual Analytics (Graduate) — Win 2023
- ECS 272 Information Visualization (Graduate) — Fall 2022
- ECS 289H AI for Visualizing Data (Graduate) — Win 2022

Teaching Assistant

- ECS 32B Introduction to Data Structures (Undergraduate) — Fall2019, Win2019
- ECS 36C Data Structures (Undergraduate) — Spring2019

Student Mentorship

- Qiaochu Wang, Ph.D. Student, City University of Hong Kong — Sep. 2025 - Present
- Yi Xu, Ph.D. Student, City University of Hong Kong — Sep. 2024 - Present
- Yunzhong Luo, Ph.D. Student, CityU & Xi'an Jiaotong University Joint Program — Sep. 2024 - Present
- Yuan Chen, Ph.D. Student, City University of Hong Kong — Sep. 2024 - June 2025
- Vincent Dorig, Undergraduate Student, ETH Zurich — Sep. 2023 - Present
- Xiwei Xuan, Ph.D. Student, UC Davis — June 2021 - Oct. 2022
- Po-Wei Chi, Master Student, UC Davis — Feb. 2021 - June. 2021

Invited Talks

- *Knowledge Organization and Visualization for Text-Dominant Multimodal Documents*
  The 13th China Visualization and Visual Analytics Conference (ChinaVis 2026) — July 2026

- *From Human to AI and Back: Knowledge and Education in the AI Era*
  The 14th International Conference on Information and Education Technology (ICIET 2026) — Apr. 2026

- *Visual Analytics for Machine Learning Model Validation*
  The 13th International Conference on Computational Visual Media (CVM 2025) Workshop — Apr. 2025

- *Harmonizing Assistance: Moderating Visual and Textual Aids in AI-Enhanced Textbook Reading*
  ETH Zurich Refresh Teaching Series Seminar — Mar. 2025

- *Use of Color in Reporting Academic Results*
  Kowloon Interaction Center (KLIC) Academic Seminar — Oct. 2024

- *Interactive Human-AI Collaboration for Knowledge Exploration and Exploitation*
  Digital Doman x CityU Faculty Symposium, City University of Hong Kong, Hong Kong — Aug. 2024

- *Fight or Friend? Unraveling the New Evolution of Education with VIS×LLM*
  Lightning Talk at 6th Workshop on Visualization for AI Explainability, VIS 2023 — Oct. 2023

- *NLP x Visualization for Interactive Knowledge Exploration and Exploitation*
  Meta x ETH AI Center Fellow Symposium, ETH Zurich — Sep. 2023

- *Visual Text Analytics*
  Guest Lecture for ECS273 Visual Analytics, University of California, Davis — Jan. 2023

- *Visual Analytics for Knowledge Exploration and Exploitation in Industry 4.0*
  VAT Valves x ETH AI Center Fellow Symposium — Aug. 2023

- *Visual Analytics for Knowledge Exploration and Exploitation*
  Lab Tour for San Joaquin Delta College Student, University of California, Davis — Nov. 2022

- *Visual Analytics for Knowledge Exploration and Exploitation*
  Center of Digital Media Computation, Xiamen University — Oct. 2022

- *Text Data Visualization*
  Guest Lecture for ECS272 Information Visualization, University of California, Davis — Oct. 2022

- *Visual Analytics for Knowledge Exploration and Exploitation*
  Envision UC Davis, University of California, Davis — Oct. 2022

- *Text Data Visualization*
  Guest Lecture for ECS289H AI for Visualizing Data, University of California, Davis — Feb. 2022

Service

Publication Chair

- ACM Designing Interactive Systems (DIS) — 2026, 2027

Program Chair

- Greater Bay ACM SIGCHI Chapter Pre-CHI Symposium 2025 — 2025

Associate Chair

- ACM CHI Conference on Human Factors in Computing Systems — 2025, 2026, 2027

Conference Session Chair

- ACM CHI Conference on Human Factors in Computing Systems — 2023, 2025

Organization Member

- AI and the Future Factory at AI House Davos 2024 — 2024
- VIS Meetup for 6th Workshop on Visualization for AI Explainability — 2023

Conference Paper Review

- IEEE VIS Conference — 2022, 2023, 2024, 2025
- ACM CHI Conference on Human Factors in Computing Systems — 2022, 2023, 2024, 2025, 2026
- ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH) — 2026
- ACM Symposium on User Interface Software and Technology (UIST) — 2026
- International Conference on Artificial Intelligence in Education (AIED)) — 2026
- ACM Conference on Intelligent User Interfaces — 2023

Invited Journal Review

- IEEE Transactions on Visualization and Computer Graphics — 2022, 2023, 2026
- ACM Transactions on Interactive Intelligent Systems — 2022

Volunteer

- Bay Area Science Festival 2022 — 2022
- Siggraph Asia 2016 — 2016

Student Club

- President of ZJU Toastmasters Speech Club, Toastmasters International — 2017

Awards

- GGCS Fall Graduate Studies Travel Award, UCDavis — Sep. 2022
- 2022 GGCS Best Graduate Researcher Award (1 out of 320), UCDavis — June 2022
- GGCSSpring Graduate Studies Travel Award, UCDavis — May 2022
- GGCS Graduate Program Fellowship 2020-21, UCDavis — Apr. 2021
- VMWare Excellent Student Fellowship — Nov. 2017
- "Graduate of Merit/Triple A graduate", Zhejiang University — Nov. 2017
- "Award of Honor for Graduate", Zhejiang University — Nov. 2017
- Academic Scholarship, Zhejiang University — Sep. 2015
- "Young Eagle Fellowship", Xiamen University — June 2015
- Second Prize of the Computer Software Design Contest of Fujian Province — Dec. 2014
- Second Prize of the National College Artificial Intelligence Design Contest — July 2014
- Meritorious Winner (First Prize) of American College Mathematical Contest in Modeling — Apr. 2014
- "Excellent Merit Student", Xiamen University — Dec. 2014
- "Jin Zhaofen & Cao Qianlong Fellowship", Xiamen University — Apr. 2013
- First Prize of the Outstanding Student Fellowship, Xiamen University — Oct. 2012